

1SAR refinement — independent re-review

protstruct_review · openscientist artefact cdba2c07-daff-4f60-ae96-12452b3a5fbb

Mostly pass · cross-tool coverage now fully closed

Ten independent oracle code paths · eight non-cctbx · all load-bearing findings $\geq$ 3-tool confirmed

- R-free reproducibly ≈ 0.21 , not 0.199 (PHENIX 0.211 · gemmi 0.217 · REFMAC 0.213)
- Round 6 R-free 0.207 better than round 7 R-free 0.213 — claim reversed
- Asn A 39 is a Ramachandran outlier (3-tool confirmed)
- Density-peak inventory incomplete — oracle 23+ / 8– at 4σ ; agent listed 9
- Mg^{2+} excluded for both ions by CheckMyMetal-style |ZI| 14–16
- Ligand / metal density support confirmed (RSCC ≥ 0.96)
- T13 data quality clean — no twinning · no tNCS · ΔB aniso 7.3 Å² (new)

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Trust model: cross-tool agreement, never PHENIX-grading-PHENIX